

* $f = 3 \text{ kHz}$
 Remember $\lambda = \frac{c}{f}$ $\begin{matrix} \swarrow \text{speed of light} \\ \nwarrow \text{frequency} \end{matrix}$

$c = 3 \times 10^8 \text{ m/s}$

$\lambda = \frac{3 \times 10^8}{3 \times 10^3} = 10^5 \text{ m} = 100 \text{ km}$

* Let's say we have $\frac{\lambda}{4}$ antenna $\Rightarrow 25 \text{ km long !!}$

* Properties of sinusoids:

$A \cdot \sin(\omega t + \phi)$

$\swarrow$ amplitude $\searrow$ angular freq (rad/s, rps) $\rightarrow$ phase (rad)

$\omega = 2\pi f$ $\searrow$ frequency (Hz = cycles/sec)

$\sin(90^\circ) \quad \sin(2)$
 $\sin(\frac{\pi}{2}) \quad \uparrow$
 2 rad.

* Amplitude Modulation (AM)

$V(t) = (V_c + V_m(t)) \cos(\omega_c t)$

$\swarrow$ carrier amplitude $\nwarrow$ signal that we want to send $\nwarrow$ carrier freq.

$\underbrace{V_c + V_m(t)}_{\text{effective amplitude}}$ $\underbrace{\cos(\omega_c t)}_{\text{carrier sinusoid}}$

AM: increase/decrease carrier amplitude according to the signal.

* Frequency Modulation (FM): $\text{total phase: } \theta$

$V(t) = V_c \cdot \cos \left\{ \omega_c t + k_f \int V_m(t) dt \right\}$

$\nwarrow$ carrier freq. $\nwarrow$ signal we want to send

$\cos(\underbrace{\omega_c t + \theta}_{\phi(t)}) \Rightarrow \frac{d\phi(t)}{dt} = \frac{d}{dt} (\omega_c t + \theta) = \omega_c$

instantaneous frequency:

$$\begin{aligned}\omega(t) &= \frac{d\theta}{dt} = \frac{d}{dt} \left\{ \omega_c t + k_f \int v_m(t) dt \right\} \\ &= \underbrace{\omega_c + k_f v_m(t)}_{\text{effective freq.}}\end{aligned}$$

Fm: increase / decrease carrier frequency according to the signal.

* Filters in frequency domain:

